

LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1
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Date: 30-08-2026

1. TITLE

Design of a College Information Portal Webpage Using HTML

2. AIM

To design a static webpage using HTML that demonstrates text formatting, ordered and unordered lists, hyperlinks, tables, images, a registration form with various input types, and embedded audio/video elements.

3. OBJECTIVE

- Learn core HTML structure and semantic container tags.
- Implement text formatting tags (, <i>, <u>, <sup>, <sub>).
- Construct ordered and unordered lists for academic departments and campus facilities.
- Build tabular representation for student academic directory.
- Design an interactive multi-field student registration form.
- Integrate multimedia elements with HTML5 <audio> and <video> tags.

4. THEORY

HTML (HyperText Markup Language) forms the foundational building block for constructing documents on the World Wide Web. Standard HTML elements structure headings, text formatting, hyperlinks, tabular listings, and interactive input forms.

Semantic elements provide structural meaning to web browsers and search engines. HTML forms collect diverse user input through text, email, password, radio buttons, checkboxes, color pickers, range controls, and select dropdowns.

5. ALGORITHM / PROCEDURE

1. Create an HTML document named task1.html.
2. Add document header with portal title and introductory description.
3. Add navigation bar linking to sections (About, Students, Registration, Media).
4. Implement formatted text demonstrating bold, italic, underline, superscript, and subscript.
5. Create ordered and unordered lists for departments and campus facilities.
6. Construct student details table with headers and data rows.
7. Design student registration form with multiple input types.
8. Embed audio and video elements with controls attribute.
9. Open in browser and verify all elements and interactions.

6. PROGRAM

task1.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>

<header>
  <h1>College Information Portal</h1>
  <p>Welcome to the College Information Portal.</p>
</header>

<nav>
  <a href="#about">About</a> |
  <a href="#students">Students</a> |
  <a href="#registration">Registration</a> |
  <a href="#media">Media</a>
</nav>

<section id="about">
  <article>

    <h2>About Our College</h2>

    <p>
      ANITS is one of the leading engineering colleges offering quality education.
    </p>

    <h3>Text Formatting</h3>

    <p>
      <b>Bold Text</b><br>
      <i>Italic Text</i><br>
      <u>Underlined Text</u><br>
      H<sub>2</sub>O (Subscript)<br>
      x<sup>2</sup> (Superscript)
    </p>

    <h3>Ordered List</h3>

    <ol>
      <li>CSE</li>
      <li>AIML</li>
      <li>ECE</li>
      <li>EEE</li>
    </ol>

    <h3>Unordered List</h3>

    <ul>
      <li>Library</li>
      <li>Laboratories</li>
      <li>Sports</li>
      <li>Hostel</li>
    </ul>

    <h3>Hyperlinks</h3>
```

```
<a href="https://anits.edu.in" target="_blank">
  College Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
  Google
</a>

</article>
</section>

<section id="students">

  <h2>Student Details</h2>

  <table border="1">

    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
      <th>Email</th>
    </tr>

    <tr>
      <td>101</td>
      <td>Bhanu</td>
      <td>AIML</td>
      <td>bhanu@gmail.com</td>
    </tr>

    <tr>
      <td>102</td>
      <td>Rahul</td>
      <td>CSE</td>
      <td>rahul@gmail.com</td>
    </tr>

    <tr>
      <td>103</td>
      <td>Priya</td>
      <td>ECE</td>
      <td>priya@gmail.com</td>
    </tr>

  </table>

  <h2>Images</h2>

  

  

</section>

<section id="registration">
  <h2>Student Registration Form</h2>

  <form>
```

```
<label>Name:</label>
<input type="text" placeholder="Enter Name">

<br><br>

<label>Email:</label>
<input type="email" placeholder="Enter Email">

<br><br>

<label>Password:</label>
<input type="password">

<br><br>

<label>Date of Birth:</label>
<input type="date">

<br><br>

<label>Age:</label>
<input type="number">

<br><br>

<label>Favourite Color:</label>
<input type="color">

<br><br>

<label>Skill Level:</label>
<input type="range" min="1" max="10">

<br><br>

<label>Gender:</label>
<br>

<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
<input type="radio" name="gender"> Other

<br><br>

<label>Hobbies:</label>
<br>

<input type="checkbox"> Reading
<input type="checkbox"> Sports
<input type="checkbox"> Music

<br><br>

<label>Branch:</label>

<select>
  <option>CSE</option>
  <option>AIML</option>
  <option>ECE</option>
  <option>EEE</option>
</select>

<br><br>

<label>Comments:</label>
```

```
<br>

<textarea rows="5" cols="40"></textarea>
<br><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">

</form>

</section>

<section id="media">

  <h2>Audio</h2>

  <audio controls>
    <source src="college.mp3" type="audio/mpeg">
    Your browser does not support audio.
  </audio>

  <h2>Video</h2>

  <video width="400" controls>
    <source src="college.mp4" type="video/mp4">
    Your browser does not support video.
  </video>

</section>

<footer>

  <p>
    &copy; 2026 College Information Portal. All Rights Reserved.
  </p>

</footer>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<h1>..<<h6>	Hierarchical headings for content structure
, <i>, <u>, <sup>, <sub>	Text formatting and scientific/mathematical notation
, ,	Ordered and unordered listing structures
<table>, <tr>, <th>, <td>	Tabular data representation for student directory
<form>, <input>, <select>	Interactive data collection form controls
<audio>, <video>	Native multimedia playback with built-in controls

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Open in browser	All sections render cleanly	Pass
TC-02	Text Formatting	Check bold, italic, sub/sup	Styles render properly	Pass
TC-03	Table Structure	Verify student directory	Grid renders with headers and rows	Pass
TC-04	Form Controls	Enter text, email, date, range, color	Controls accept inputs	Pass
TC-05	Media Playback	Inspect audio and video controls	Media player widgets appear	Pass

9. OUTPUT

Output 1: Execution Screenshot

College Information Portal

Welcome to the College Information Portal.

[About](#) | [Students](#) | [Registration](#) | [Media](#)

About Our College

ANITS is one of the leading engineering colleges offering quality education.

Text Formatting

Bold Text

Italic Text

Underlined Text

H₂O (Subscript)

x² (Superscript)

Ordered List

1. CSE

2. AIML

3. ECE

4. EEE

Unordered List

- Library

- Laboratories

- Sports

- Hostel

Hyperlinks

[College Website](#)

[Google](#)

Student Details

Roll No	Name	Branch	Email
---------	------	--------	-------

101	Bhanu	AIML	bhanu@gmail.com
-----	-------	------	-----------------

102	Rahul	CSE	rahul@gmail.com
-----	-------	-----	-----------------

103	Priya	ECE	priya@gmail.com
-----	-------	-----	-----------------

Images



10. RESULT

Thus, the experiment for Observation Task 1 (Design of a College Information Portal Webpage Using HTML) was successfully executed and verified.